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EXPERIMENT NO. 7

CONFIGURATION OF LINK STATE ROUTING PROTOCOL (OSPF)

AIM:

To configure and analyze OSPF routing protocol using simulation tools.

OBJECTIVES

- To configure OSPF areas and routers.
- To understand Link State Advertisements.
- To analyze the Shortest Path First algorithm.
- To observe routing convergence.
- To verify routing using simulation traffic.

THEORY

Open Shortest Path First is a link state routing protocol used within an autonomous system. It builds a topology database and calculates the shortest path using Dijkstra algorithm.

OSPF uses cost as its metric and supports hierarchical routes through areas. It converges faster than RIP.

SOFTWARE / MATERIALS REQUIRED

- Cisco Packet Tracer 8.x
- Computer System

IMPORTANT COMMANDS / PROGRAM STEPS

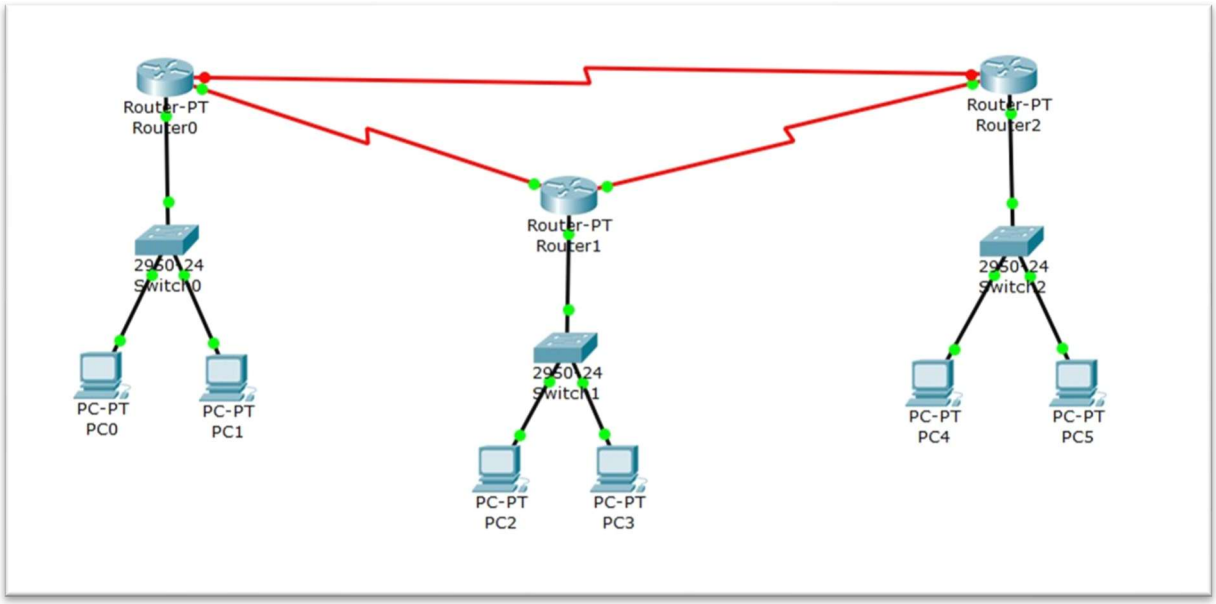
- router ospf 1
- network 192.168.1.0 0.0.0.255 area 0
- show ip ospf neighbor
- show ip route
- show ip ospf database
- ping 192.168.2.2

PROCEDURE

- Assign IP addresses to routers and PCs.
- Enable OSPF process on each router.
- Advertise networks with wildcard masks and area number.
- Verify neighbor formation using show ip ospf neighbor.
- Check routing table and OSPF database.
- Test connectivity using ping.

OUTPUT:

When Wan Link between Router0 & Router2 is inactive. (Before applying OSPF Protocol):




```
        --- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router0
Router0(config)#in fa0/0
Router0(config-if)#ip addr 10.0.0.1 255.0.0.0
Router0(config-if)#no shut

Router0(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router0(config)#in se2/0
Router0(config-if)#ip addr 172.16.0.1 255.255.0.0
Router0(config-if)#clock rate 64000
Router0(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router0(config-if)#in se3/0
Router0(config-if)#ip addr 172.20.0.1 255.255.0.0
Router0(config-if)#^Z
Router0#
%SYS-5-CONFIG_I: Configured from console by console

Router0#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router1
Router1(config)#in fa0/0
Router1(config-if)#ip addr 20.0.0.1 255.0.0.0
Router1(config-if)#no shut

Router1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router1(config)#in se2/0
Router1(config-if)#ip addr 172.16.0.2 255.255.0.0
Router1(config-if)#no shut

Router1(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
exit
Router1(config)#in se
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state t
Router1(config)#in se3/0
Router1(config-if)#ip addr 172.18.0.1 255.255.0.0
Router1(config-if)#clock rate 64000
Router1(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router1(config-if)#^Z
Router1#
%SYS-5-CONFIG_I: Configured from console by console

Router1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    20.0.0.0/8 is directly connected, FastEthernet0/0
```

```
        --- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router2
Router2(config)#in fa0/0
Router2(config-if)#ip addr 30.0.0.1 255.0.0.0
Router2(config-if)#no shut

Router2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router2(config)#in se2/0
Router2(config-if)#ip addr 172.18.0.2 255.255.0.0
Router2(config-if)#no shut

Router2(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

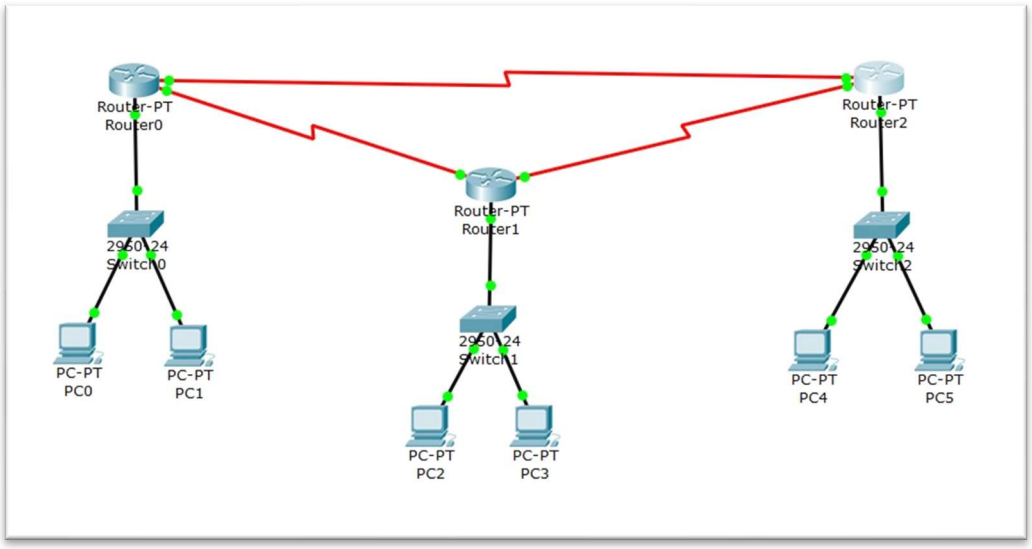
Router2(config-if)#in se3/0
Router2(config-if)#ip addr 172.20.0.2 255.255.0.0
Router2(config-if)#^Z
Router2#
%SYS-5-CONFIG_I: Configured from console by console

Router2#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    30.0.0.0/8 is directly connected, FastEthernet0/0
```

When WAN Link between Router0 & Router2 is active. (After applying OSPF Protocol):



Router0 Commands:

```
Router0>en
Router0#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
O    20.0.0.0/8 [110/21] via 172.16.0.2, 01:01:16, Serial2/0
O    30.0.0.0/8 [110/14] via 172.20.0.2, 00:53:42, Serial3/0
C    172.16.0.0/16 is directly connected, Serial2/0
O    172.18.0.0/16 [110/77] via 172.20.0.2, 00:53:42, Serial3/0
C    172.20.0.0/16 is directly connected, Serial3/0
Router0#sh ip route ospf
O    20.0.0.0 [110/21] via 172.16.0.2, 01:01:25, Serial2/0
O    30.0.0.0 [110/14] via 172.20.0.2, 00:53:51, Serial3/0
O    172.18.0.0 [110/77] via 172.20.0.2, 00:53:51, Serial3/0
Router0#sh ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
172.18.0.1        0   FULL/  -        00:00:34    172.16.0.2      Serial2/0
172.18.0.2        0   FULL/  -        00:00:32    172.20.0.2      Serial3/0
Router0#sh ip ospf database
            OSPF Router with ID (172.16.0.1) (Process ID 10)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
172.16.0.1      172.16.0.1      1426        0x80000007 0x00c408 5
172.18.0.1      172.18.0.1      1579        0x80000007 0x002510 5
172.18.0.2      172.18.0.2      1425        0x80000006 0x00bd34 5
Router0#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router0(config)#in se3/0
Router0(config-if)#no shut
Router0(config-if)#^Z
Router0#
%SYS-5-CONFIG_I: Configured from console by console

Router0#ping 30.0.0.0

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 30.0.0.0, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/10/18 ms
```

```
Router0#traceroute 30.0.0.0
Type escape sequence to abort.
Tracing the route to 30.0.0.0

  1   172.20.0.2      1 msec    0 msec    0 msec
  2   *     *     *
  3   *     *     *
  4   *     *     *
  5   *     *     *
  6   *     *     *
```

Router1 Commands:

```
Router1>en
Router1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router1(config)#router ospf 100
Router1(config-router)#network 20.0.0.0 0.255.255.255 area 0
Router1(config-router)#network 172.16.0.0 0.0.255.255 area 0
Router1(config-router)#network 172.18.0.0 0.0.255.255 area 0

Router1(config-router)#in se3/0
Router1(config-if)#bandwidth 2500

Router1>en
Router1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

O    10.0.0.0/8 [110/65] via 172.16.0.1, 00:48:02, Serial2/0
C    20.0.0.0/8 is directly connected, FastEthernet0/0
O    30.0.0.0/8 [110/41] via 172.18.0.2, 00:43:04, Serial3/0
C    172.16.0.0/16 is directly connected, Serial2/0
C    172.18.0.0/16 is directly connected, Serial3/0
O    172.20.0.0/16 [110/104] via 172.18.0.2, 00:40:30, Serial3/0
Router1#sh ip route ospf
O    10.0.0.0 [110/65] via 172.16.0.1, 00:48:11, Serial2/0
O    30.0.0.0 [110/41] via 172.18.0.2, 00:43:13, Serial3/0
O    172.20.0.0 [110/104] via 172.18.0.2, 00:40:39, Serial3/0
Router1#sh ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
172.16.0.1        0   FULL/  -        00:00:31    172.16.0.1      Serial2/0
172.18.0.2        0   FULL/  -        00:00:33    172.18.0.2      Serial3/0
Router1#sh ip ospf database
            OSPF Router with ID (172.18.0.1) (Process ID 100)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
172.18.0.1      172.18.0.1      798         0x80000007 0x002510 5
172.16.0.1      172.16.0.1      645         0x80000007 0x00c408 5
172.18.0.2      172.18.0.2      644         0x80000006 0x00bd34 5
```

```
Router1>en
Router1#ping 10.0.0.0

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.0, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/9/17 ms

Router1#ping 30.0.0.0

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 30.0.0.0, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/3/7 ms
```

Router2 Commands:

```
Router2>en
Router2#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

O    10.0.0.0/8 [110/65] via 172.20.0.1, 01:01:23, Serial3/0
O    20.0.0.0/8 [110/65] via 172.18.0.1, 01:03:55, Serial2/0
C    30.0.0.0/8 is directly connected, FastEthernet0/0
O    172.16.0.0/16 [110/128] via 172.18.0.1, 01:03:55, Serial2/0
C    172.18.0.0/16 is directly connected, Serial2/0
C    172.20.0.0/16 is directly connected, Serial3/0
Router2#sh ip route ospf
O    10.0.0.0 [110/65] via 172.20.0.1, 01:01:26, Serial3/0
O    20.0.0.0 [110/65] via 172.18.0.1, 01:03:59, Serial2/0
O    172.16.0.0 [110/128] via 172.18.0.1, 01:03:59, Serial2/0
Router2#sh ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
172.16.0.1        0   FULL/  -        00:00:34    172.20.0.1      Serial3/0
172.18.0.1        0   FULL/  -        00:00:30    172.18.0.1      Serial2/0
Router2#sh ip ospf database
            OSPF Router with ID (172.18.0.2) (Process ID 1000)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
172.18.0.2      172.18.0.2      72          0x80000007 0x00bb35 5
172.18.0.1      172.18.0.1      228         0x80000008 0x002311 5
172.16.0.1      172.16.0.1      74          0x80000008 0x00c209 5
Router2#ping 10.0.0.0

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.0, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/9/17 ms
```

```
Router2#traceroute 10.0.0.0
Type escape sequence to abort.
Tracing the route to 10.0.0.0

  1   172.20.0.1      14 msec   2 msec    0 msec
  2   *     *     *
  3   *     *     *
  4   *     *     *
  5   *     *     *
  6   *     *     *
```

CONCLUSION:

The OSPF routing protocol was successfully configured and tested using Cisco Packet Tracer. All the routers formed neighbor relationships and shared information about their connected networks. Based on this information, OSPF selected the shortest and most efficient routes using the SPF algorithm. The routes learned through OSPF were identified by the letter O in the routing table. Successful ping results confirmed that devices on different networks could communicate with each other. Therefore, this experiment helped us understand the working, route selection, and fast convergence of the OSPF routing protocol.